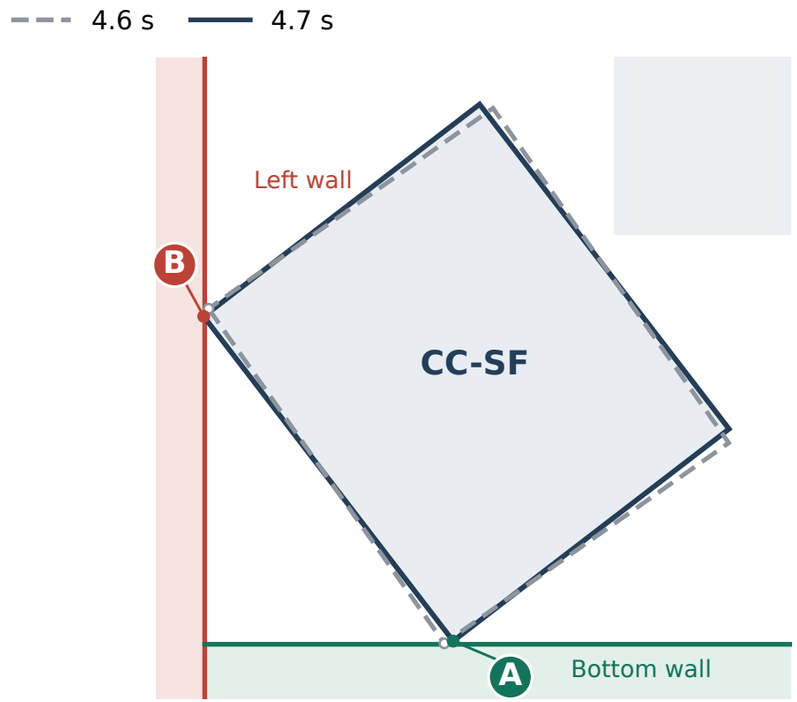
More clearance to one wall, penetration of the other — in 0.1 s

DoN 0.85 · CC-SF seed 10 · actual recorded poses at 4.6 and 4.7 s

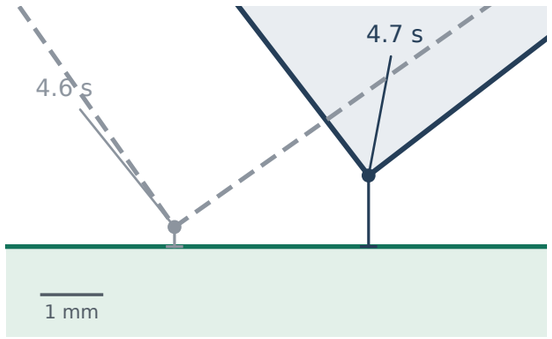
Actual footprint and wall contacts



Center displacement 1.42 mm · rotation +2.37°

A Bottom wall

Previously closest: gap increases



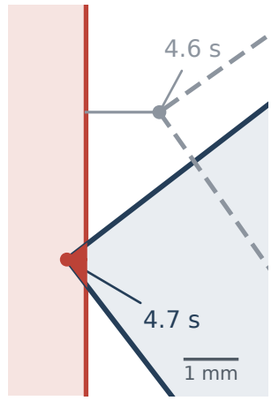
Magnified contact · actual geometry

0.326 → 1.182 mm

Gap +0.856 mm

B Left wall

Other nearby wall: penetration



Red area = actual penetration

1.408 → -0.368 mm

0.368 mm penetration

Old-wall gap improves; another guard is crossed

Other-wall distance < 0.05 mm · all 50 matched seeds

CC-SF 17/50

34% of trials

CC-MF 2/50

4% of trials

Two actual recorded poses; no exaggerated motion or rotation. Only the contact views are magnified. Observed motion does not recover the optimizer’s internal feature choice.